

System Design Document

Team 1

Table of Contents

Conceptual System Design	pg. 2
Report formats.....	pg. 2
Screen layouts/screen shots.....	pg. 4
Technical Design, high level	pg. 5
Detailed class diagrams.....	pg. 5
Supporting text specification.....	pg. 6
Database table descriptions.....	pg. 11
Technical support specification.....	pg. 15

Conceptual System Design

Report formats

Weekly Employee Activity Sample Report

Employee Name	Position	Monday Hours	Tuesday Hours	Wednesday Hours	Thursday Hours	Friday Hours	Saturday Hours	Sunday Hours	Total Hours Worked	Tasks Completed
Sarah Thomas	Waiter	8	8	6	8	0	7	0	37	52 orders served and completed
Emma Johnson	Waiter	0	7	8	8	6	7	0	36	61 orders served and completed
Nicholas Roberts	Waiter	6	6	0	7	6	5	0	30	35 orders served and completed
Jane Banks	Cook	8	8	8	8	8	9	0	49	104 meals prepared
Theodore Daniels	Cook	7	8	0	8	8	8	0	39	89 meals prepared
Blake Carlsen	Cook	0	7	7	7	6	6	0	33	62 meals prepared
Doug Cox	Host	7	8	8	8	7	7	0	45	63 table assignments made
Kathleen White	Host	0	6	6	6	6	5	0	29	33 table assignments made
Tom Padgett	Busboy	6	6	6	6	6	7	0	37	51 tables cleaned
Richard Stone	Busboy	0	5	6	0	5	6	0	22	32 tables cleaned
Dean Gaunt	Busboy	6	7	6	6	6	6	0	37	44 tables cleaned
Matthew Brown	Manager	9	9	9	9	9	9	0	54	12 changes made to restaurant operations

Weekly Revenue Sample Report

Day	Total Revenue	#	Total Orders	Most Popular Menu Item	Least Popular Menu Item
Monday	\$2,140		85	Grilled Pork Chops	Wedge Salad
Tuesday	\$2,460		92	New York Strip Steak	Catfish Bites
Wednesday	\$2,300		90	Shrimp & Grits	Fried Veggies
Thursday	\$2,780		103	Seared Tuna	House Salad
Friday	\$4,550		162	New York Strip Steak	Mac & Cheese
Saturday	\$5,100		180	Caribbean Chicken	Club Sandwich
Sunday	\$3,970		149	Grilled Pork Chops	Catfish Bites
Total	\$23,300		861		

Weekly Efficiency and Traffic Sample Report

Data	Data Entry
Average Preparation Time	13 min.
Average Turnaround Time	47 min.
Highest Traffic Period	Friday 6-8:30 pm
Lowest Traffic Period	Monday 2:30-4 pm

Weekly Receipt Sample Report

#	Order Number	Waiter	#	Table Number	Time	Payment Method	Subtotal	Tip	Total
	1	Emma Johnson		4	11:15 AM	Credit Card	\$42.50	\$7.00	\$49.50
	2	Nicholas Roberts		7	11:40 AM	Cash	\$38.00	\$5.00	\$43.00
	3	Sarah Thomas		10	12:05 PM	Credit Card	\$56.25	\$10.00	\$66.25
	4	Nicholas Roberts		2	12:35 PM	Debit Card	\$31.00	\$6.00	\$37.00
	5	Emma Johnson		8	1:10 PM	Credit Card	\$48.90	\$7.00	\$55.90
	6	Sarah Thomas		1	1:45 PM	Credit Card	\$60.00	\$8.00	\$68.00
	7	Nicholas Roberts		5	2:30 PM	Cash	\$27.00	\$3.00	\$30.00
	8	Emma Johnson		10	3:15 PM	Debit Card	\$35.75	\$6.25	\$42.00
	9	Nicholas Roberts		6	4:00 PM	Credit Card	\$44.20	\$5.80	\$50.00
	10	Emma Johnson		8	5:05 PM	Credit Card	\$59.00	\$9.00	\$68.00
	11	Sarah Thomas		4	6:20 PM	Debit Card	\$46.10	\$6.90	\$53.00
	12	Sarah Thomas		3	7:00 PM	Credit Card	\$52.50	\$7.50	\$60.00
	13	Nicholas Roberts		7	8:10 PM	Credit Card	\$63.00	\$10.00	\$73.00
	14	Emma Johnson		9	9:05 PM	Cash	\$28.00	\$5.00	\$33.00

Screen layouts/screen shots

Welcome to J's Restaurant

[Shutdown Button]

Please Enter Login Information

Username

Password

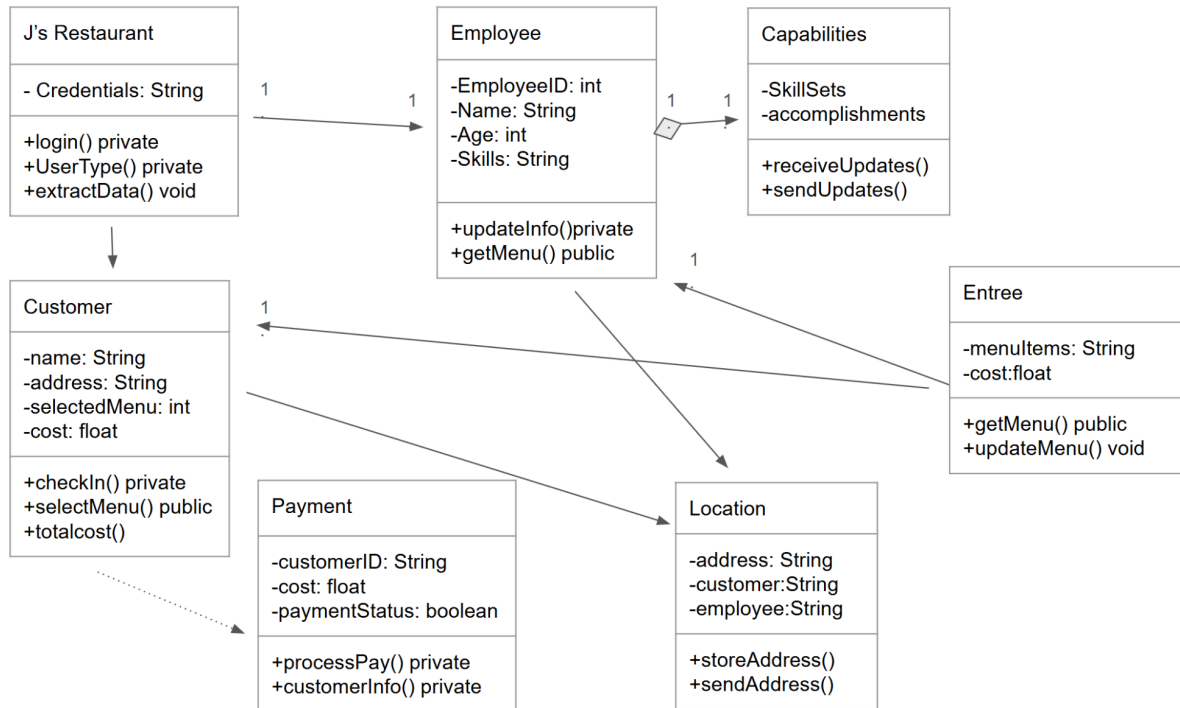
Clear All Submit

Log-in Screen
All employees will need to log into the system
Use Cases: 1.2, 2.2, 3.2, 4.2, 5.2

[Click here to view all screen layouts](#)

Technical Design, high level

Detailed class diagrams



Supporting text specification

Public class JRestaurant

Constructor	Description	
String()	Initializes the JResturant class with the given credentials	
Field Summary		
Modifier and Type	Field	Description
String	credentials	Stores user credentials for authentication
Method Summary		
Modifier and Type	Field	Description
Boolean	login(username, password)	Allows access to the user based on credentials
String	UserType()	Determines whether the user is a customer or Employee
void	extractData()	Extract information form either the Employee or Customer classes

Public class Employee

Constructor	Description
int	Initializes an employee with an ID, name and address. Also creates and empty lists for skill sets and

	accomplishments	
Field Summary		
Modifier and Type	Field	Description
int	employeeID	Identifiers for employees
String	name	Name of employee
String	address	Contains the employee’s address retrieved from the location class
List<String>	skillsets	Skill sets of employee
List<String>	accomplishments	Tracks daily accomplishments of employee
Method Summary		
Modifier and Type	Field	Description
void	updateInfo()	Updates employee information when needed
List<String>	getMenu()	Retrieves information from the Entree class

Public Class Capability

Constructor	Description	
ArrayList<>	Initializes empty lists for skill sets and accomplishments	
Field Summary		
Modifier and Type	Field	Description
List<String>	skillsets	Skill sets for employees
List<String>	accomplishments	Daily accomplishments for employees
Method Summary		

Modifier and Type	Field	Description
String	recieveUpdates()	Allows access to the user based on credentials
List<String>	sendUpdates()	Sends new data back to the Employee class

Public class Customer

Constructor	Description	
String()	Initializes a customer with an ID, name, and address. Default values are set for selected menu and cost	
Field Summary		
Modifier and Type	Field	Description
int	customerID	Identifier for customer
String	name	Name for customer
String	address	Retrieves customer's address from location class
String	selectedMenu	Menu item chosen by customer
float	cost	Cost of the chosen menu item
Method Summary		
Modifier and Type	Field	Description
void	checkIn()	Registers the customer as present in the restaurant
String	selectMenu(List<String> Menu)	Allows the customer to select an item from the menu
void	pay(float cost)	Sends payment information to the payment class

Public class Payment

Constructor		Description
boolean		Initializes a payment with an ID and cost. Sets payment status to false.
Field Summary		
Modifier and Type	Field	Description
int	CustomerID	Retrieve customer ID from the customer class
float	cost	Amount needed to be paid
boolean	PaymentStatus	Tracks whether the payment is completed
Method Summary		
Modifier and Type	Field	Description
void	processPay(float amount)	Process the customer's payment and marks it as completed
void	customerInfo()	Sotres payment details for future retrieval by other classes.

Public class Location

Constructor		Description
String()		Initializes empty lists for employee and customer addresses

Field Summary		
Modifier and Type	Field	Description
List<String>	employeeAddress	Store addresses for employee by ID
List<String>	customerAddress	Store addresses for customer by ID
Method Summary		
Modifier and Type	Field	Description
String	storeAddress(String address, int ID)	Store address for employee or customer based on ID
int	sendAddress(int ID)	Retrieves the address information for a given ID

Public class Entree

Constructor	Description	
ArrayList<>()	Initialized empty lists for menu items	
Field Summary		
Modifier and Type	Field	Description
List<String>	menuItems	List of all menu items
float	cost	Stores cost information for menu item.
Method Summary		
Modifier and Type	Field	Description
List<String>	getMenu()	Allows access to the user based on credentials
void	updateMenu()	Determines whether the user is a customer or Employee

Database table descriptions

Payment			
Key	Field Name	Data Type	Tt Description
PK	Payment ID	Integer(16) Not Null	The unique (surrogate) key for each transaction.
	Payment Type	Char(25)	The payment type or method.
	Payment Amount	Money	The amount of money that is paid or is about to be paid.

Location			
Key	Field Name	Data Type	Tt Description
PK	Street Num	Varchar(50) Not Null	The unique (surrogate) key for each Location added into the database..
	ZIP	Char(10)	The ZIP code.
	Address	Varchar(50)	The address.
	State	Char(2)	The state.
	City	Varchar(50)	The city.

Customer

Key	Field Name	Data Type	Tt Description
PK	Customer ID	Integer Not Null	The unique (surrogate) key for each customer.
	FName	Varchar(35)	Customer's first name.
	LName	Varchar(35)	Customer's last name.
	Phone #	Char(13) Unique	The Customer's phone number.
	DOB	Date	The customer's date of birth.
	Age	Derived From DOB	The age calculated from their date of birth.

Employee

Key	Field Name	Data Type	Tt Description
PK	Employee ID	Integer Not Null	The unique (surrogate) key for each employee.
	FName	Varchar(35)	Employee's first name.
	LName	Varchar(35)	Employee's last name.
	Phone Num	Char(13) Unique	The employee's phone number.
	DOB	Date	The employee's date of birth.
	Age	Derived From DOB	The age calculated from their date of birth.
	Position	Varchar(20)	The employee's assigned position.
	Salary	Money	The employee's set salary per hour.
	Activity Info	Varchar	A description of what an employee did during their shift.

Capabilities			
Key	Field Name	Data Type	T Description
PK	Capabilities ID	Integer Not Null	The unique (surrogate) key for each employee's skills.
	Description	Varchar(50)	A description explaining the employee's set of skills .

J's Restaurant			
Key	Field Name	Data Type	T Description
PK	Store ID	Integer Not Null	The unique (surrogate) key for each J's Restaurant.
	Phone #	Char(13) Unique	The restaurant's specific phone number.
	Hours	Time	The time when the restaurant is open or will close on certain days.
	Revenue	Money	How much money the restaurant profited or lost in a day.

Entree			
Key	Field Name	Data Type	Description
PK	Entree ID	Integer Not Null	The unique (surrogate) key for each entree dish.
WK	Employee ID	Integer Not Null	The associative key from an employee who is assigned to the entree.
WK	Store ID	Integer Not Null	The associative key from the restaurant that is assigned to the entree.
WK	Customer ID	Integer Not Null	The associative key from a customer who ordered the entree.
	Price	Money	The cost of the entree.
	Ingredients	Varchar(50)	Ingredients used in the entree.
	Prep Time	Time	The estimated time that the entree will take to be prepared.

Relationships:

- Customer uses Payment in order to pay for their entree
- Customer orders and waits for Entree to be served
- Entree is served by Employee and is ordered at J's Restaurant
- Employee works for J's Restaurant
- J's Restaurant hosts the Entree for the customer
- The Employee has a set of capabilities

Technical support specification

Introduction

With the program being complete and tested, it's time to bring it into J's Restaurant. As this is a big change for the restaurant, we plan to provide multiple forms of support to help this transition go as smoothly as possible. From training employees to helping months after the program has been brought in, we have different approaches. This document will explain said approaches.

Installation Support

During the installation process, our staff will come to the restaurant and install the terminals in each area of the restaurant. These areas include on the floor for waiters and busboys, in the kitchen for the chef, at the front of the restaurant for the host, and in the back of the restaurant for managers. After the terminals have been installed, our staff will test the functionality of the program. Users will also be asked to test the program. Our project manager will then run the program with the owner/head manager of the restaurant and answer any questions. Once test running has been completed, we will provide the owner/head manager with a manual in preparation for Training Day.

Training Approach

To get employees familiar with the new system, we plan to train employees (managers, hosts, waiters, chefs, and busboys) on the program by holding a "Training Day". Training Day will be held on a Sunday, when the restaurant is not in operation. During Training Day, we will introduce the program as a whole to employees. Employees will then be grouped based on their positions and given booklets that will walk them through the screens available to them on the program. Next, employees will be asked to go to their respective areas (e.g. host at the entrance and waiters at the stationary terminals) in the restaurant to practice using the programs themselves. At each station, there will be someone to assist employees and answer any questions they may have. Lastly, employees will be asked to get into groups to play a game of Jeopardy led by our staff. The questions will be on the new program. This is a great way to clear any misconceptions on the program and answer any questions that arise.

Training Materials

During Training Day, employees will be provided with materials to help them get familiar with the new system. Along with the PowerPoint Presentation displayed at the beginning of Training Day, employees will also be provided with booklets. There will be five different types of booklets, each one specific to a certain employee type (manager, host, waiter, chef, and busboy). Each employee will get a booklet respective of their position. The booklets will give a brief introduction to the system and walk through each screen available to that employee. Each booklet will also include a QR code with a link to a video demonstration specific to each position's usage. The owner/head manager will also be provided with the manual for the program. The PowerPoint, booklets, and manuals are all materials that will be provided in the hopes of solving any issues had with the system.

Onsite/Remote Assistance

If the system is malfunctioning or employees need help of any sort, they should call the help desk number provided to them in the manual. This way, they can receive remote assistance. However, if the issue is too complex to be solved remotely, one of our staff can report on-site within 24 hours. During the wait, we ask employees not to use the terminals to prevent further damage.

Updates

The system will have major updates yearly, but minor updates every six months. When it is time for an update, the user will be given the option to select a date and time for the system to update. If employees opt not to choose a time, the system will automatically update at midnight. After each major update (each year), employees will be provided with a video demonstration of new features and any changes made to existing features.

Troubleshooting

If we are contacted with an issue, our help desk staff will first ask the employee to describe the issue they are running into. From there, the following troubleshooting steps will be followed:

1. Establish a theory
2. Establish action plan
3. Implement action plan
4. Verify full functionality
5. Document issue, solution, and outcome

Once the problem is identified, our staff will then establish a theory on what the possible issue could be. This step can include identifying the scope of the issue or looking at similar problems and their solutions. Once we have a theory, we can move on to creating an action plan. This will mostly be done on our part. Next, we will implement the action plan by asking employees to follow directions. If the assistance is remote, this step will consist of a lot of communication between our staff and the employee calling. Once the plan is implemented, our staff will confirm the system is working properly. If it is not, we will return to step 1. If it is, we will document this issue and its solution for future reference.

Help Desk Hours

If employees need any assistance, our help desk can be contacted at (123)456-7891 during these hours:

Monday - Friday: 9:00 a.m. - 10:00 p.m.

Saturday: 9:00 a.m. - 7:00 p.m.

Sunday: 10:00 a.m. - 5:00 p.m.